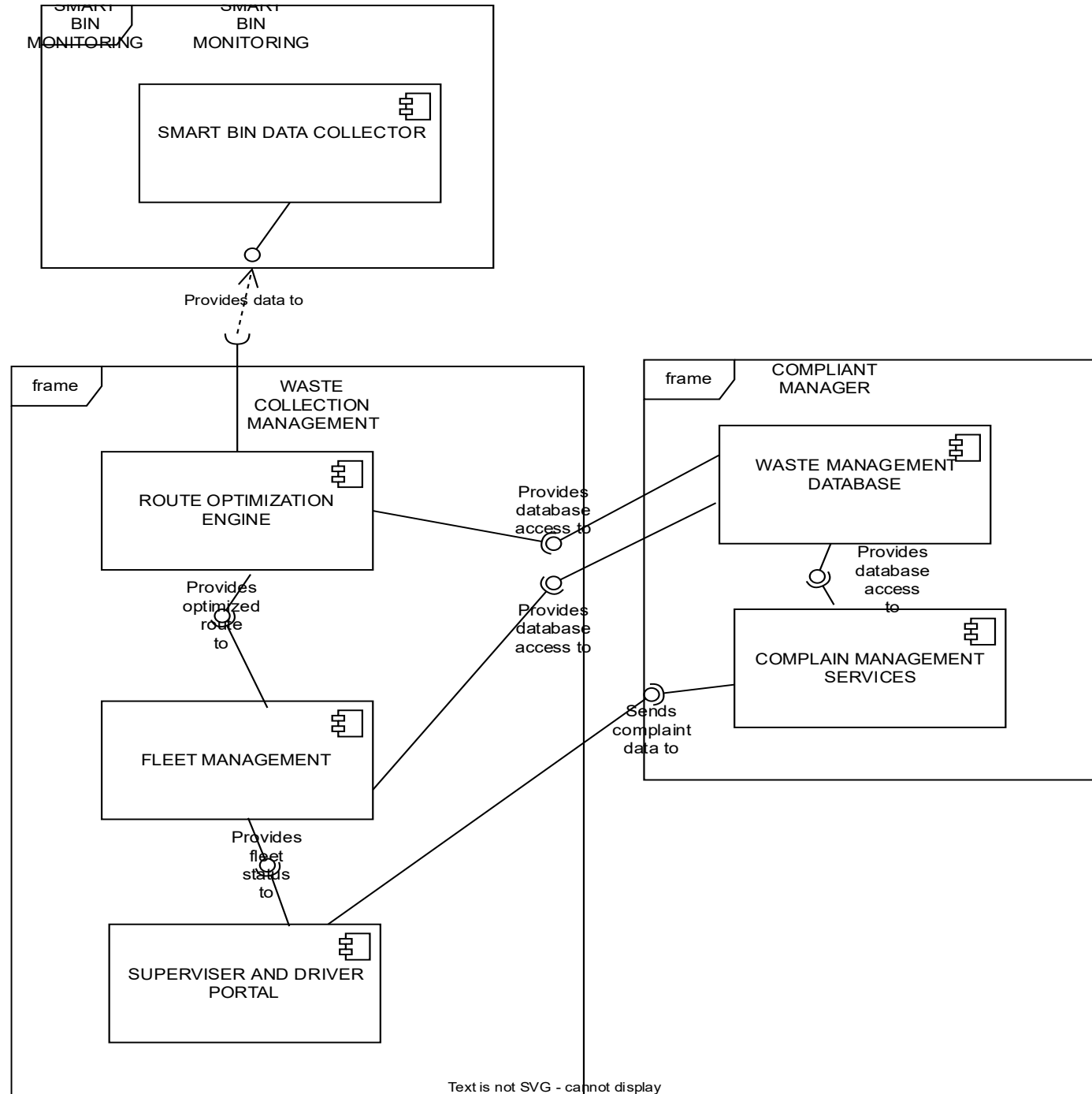


SE LAB 03

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ARCHITECTURE STYLE JUSTIFICATION

1. Justification for Component-Based Architecture (CBA)

- **Decoupling and Independent Evolution:** The system decomposes distinct operational responsibilities—such as telemetry collection (SMART BIN DATA COLLECTOR), routing logistics (ROUTE OPTIMIZATION ENGINE), and complaint tracking (COMPLAIN MANAGEMENT SERVICES)—into autonomous components. Modifying the route optimization algorithm or swapping the complaint management engine can be executed independently without impacting the rest of the application structure.
- **Encapsulation & Contract Enforcement:** Components communicate strictly via explicit provided/required interfaces (lollipop and socket notation). Internal implementations remain hidden, ensuring that components depend only on functional contracts rather than underlying code details.

2. Justification for Repository / Shared Data Architecture

- **Centralized Data Consistency:** Core operations across different subsystems—such as calculating optimal routes, tracking fleet state, and logging complaints—require a single source of truth for persistent entity data. Utilizing a centralized WASTE MANAGEMENT DATABASE component ensures data integrity and eliminates state duplication across processing modules.
- **Shared Access Abstraction:** The database component exposes explicit database access interfaces (*"Provides database access to"*), enabling components like the ROUTE OPTIMIZATION ENGINE, FLEET MANAGEMENT, and COMPLAIN MANAGEMENT SERVICES to query and persist state uniformly.

3. Justification for Layered / Subsystem Domain Partitioning

- **Clear Separation of Concerns (SoC):** The architecture groups logically related components into distinct high-level UML frame boundaries:
 - **Smart Bin Monitoring:** Isolates edge telemetry data collection from core business logic.
 - **Waste Collection Management:** Consolidates operational and execution workflows (routing, fleet tracking, portal interaction).
 - **Compliant Manager:** Dedicates resources specifically to governance, complaint handling, and persistent data operations.
- **Maintainability & Scalability:** Grouping responsibilities into explicit subsystems allows development teams to work in parallel on isolated modules and allows individual subsystems to scale based on load (e.g., scaling data collection during high sensor ingestion without scaling the driver portal).